

Course Title: Technology, Environment and Society

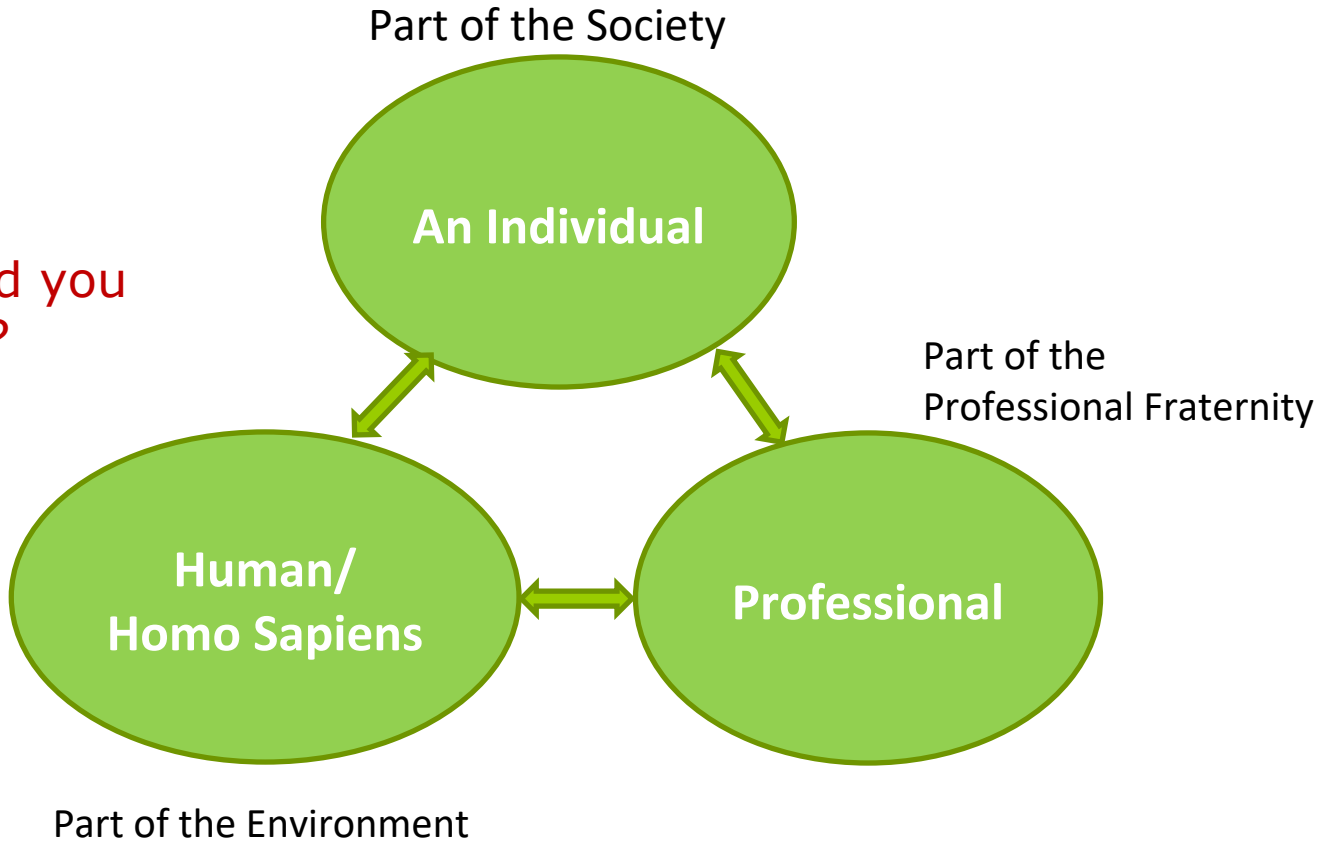
HUM 4147

Topic 1: Environment and Ecology

- Relationship among Environment, Society and Technology
- Environment and Earth System
- Ecology and Ecosystem
- Structure and Function of Ecosystem
  - ✓ Energy Flow
- Carrying Capacity of Natural Resources

Relationship among Environment, Society and Technology

How would you explain???



Relationship among Environment, Society and Technology

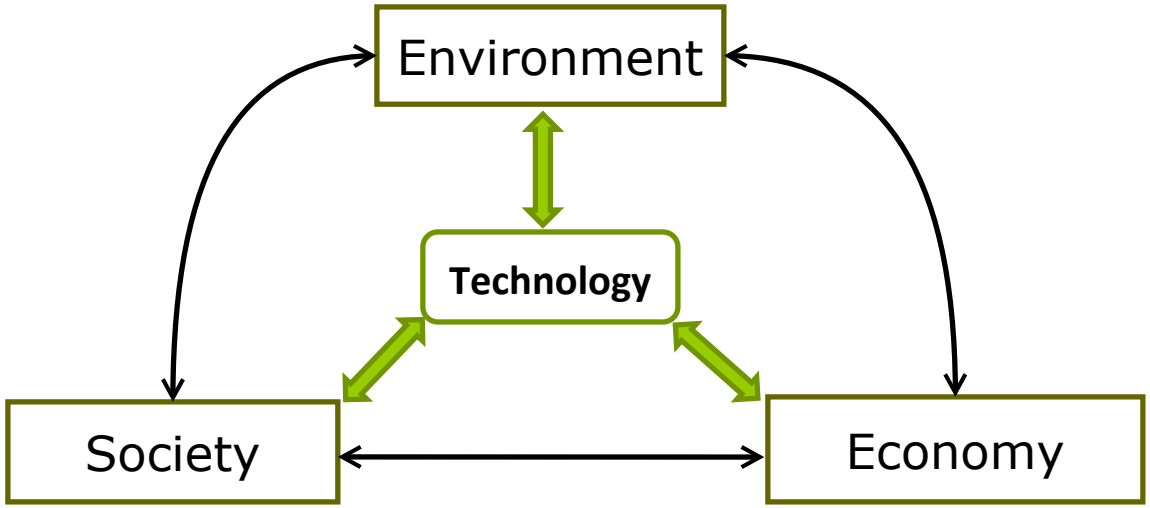
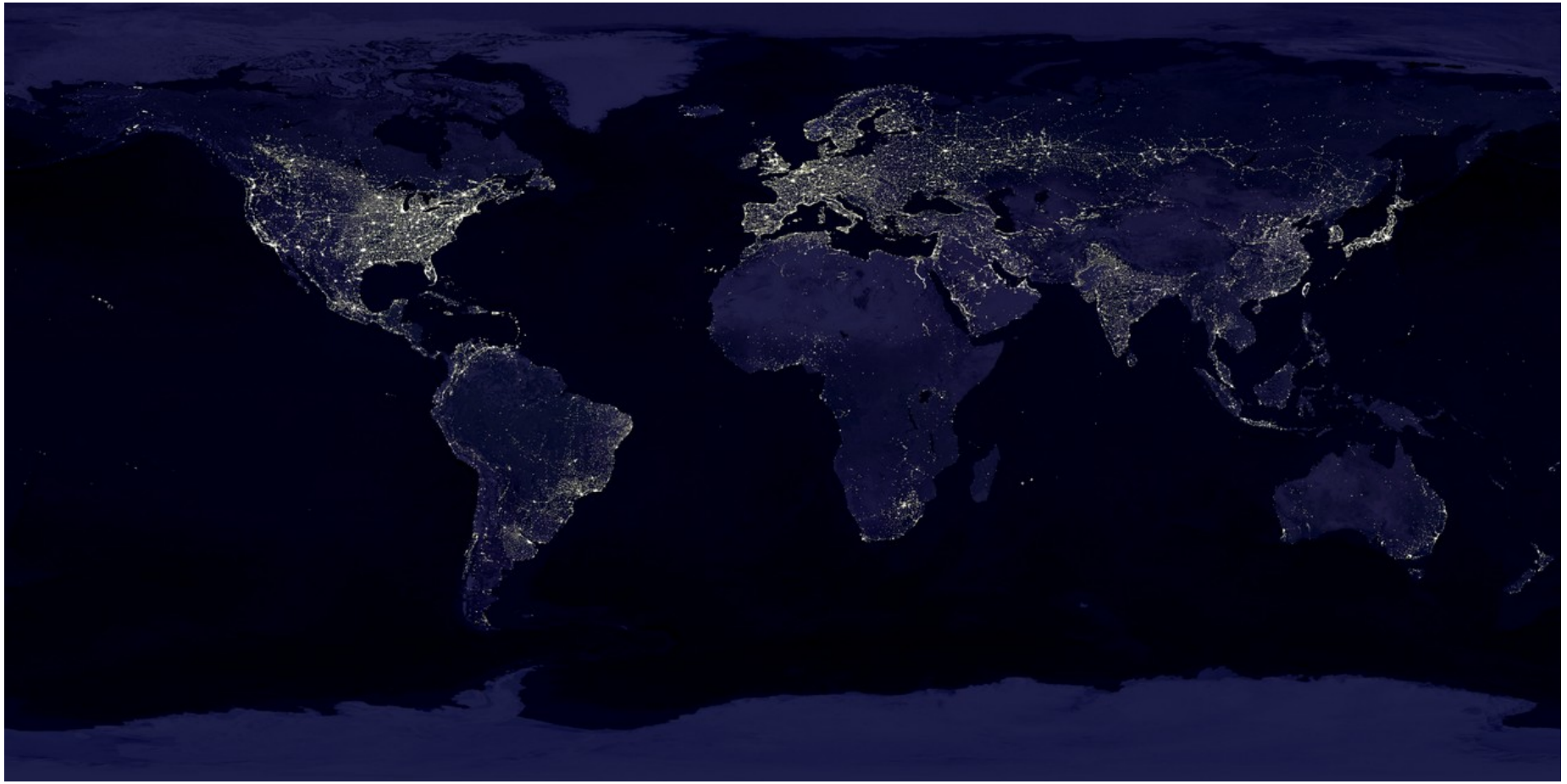


Figure: Dimension of Sustainable Triangle

Relationship among Environment, Society and Technology

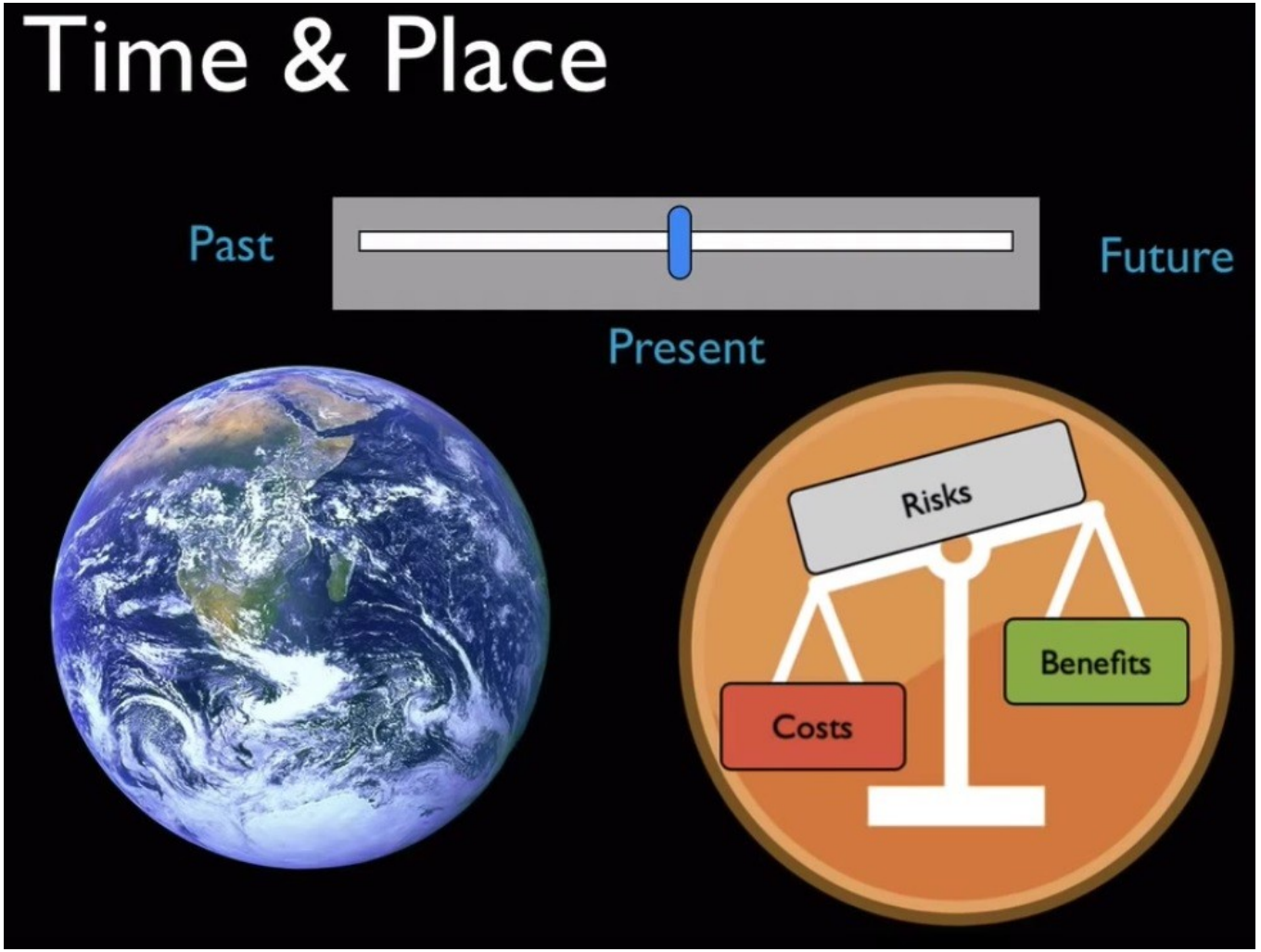


NASA Image, 2012

Relationship among Environment, Society and Technology



Relationship among Environment, Society and Technology



Environment

The environment is the aggregate of all external factors or conditions that influence the activities and existence of all living beings including human.

Or

the environment is the interaction within and among physical, living and human world in our **surrounding.**

Surroundings!!!

- A certain distance?
- the room; we live in?
- building?
- City?
- District or division?
- Country?
- Planet?

Environment and Earth System

Earth can be divided into three different compartments or world:

1. Physical world/Physical; consists of non living objects (e.g. Rocks, soil, cloud, air) and processes (e.g. Evaporation, season, weather, climate etc.).

2. Living world/Biological; consists of living things (e.g. Millions of species of plants, animals, fungus and bacteria's) and life process (e.g. feeding, reproduction, respiration and photo-synthesis etc.)

3. Human world/Social; consists of all things that human make, do and believe (e.g. science, arts, religion, literature etc.)

Environment and Earth System

Do these three 'worlds' exist in total isolation from one another?

Of course not!!!

Interaction between PW, LW and HW

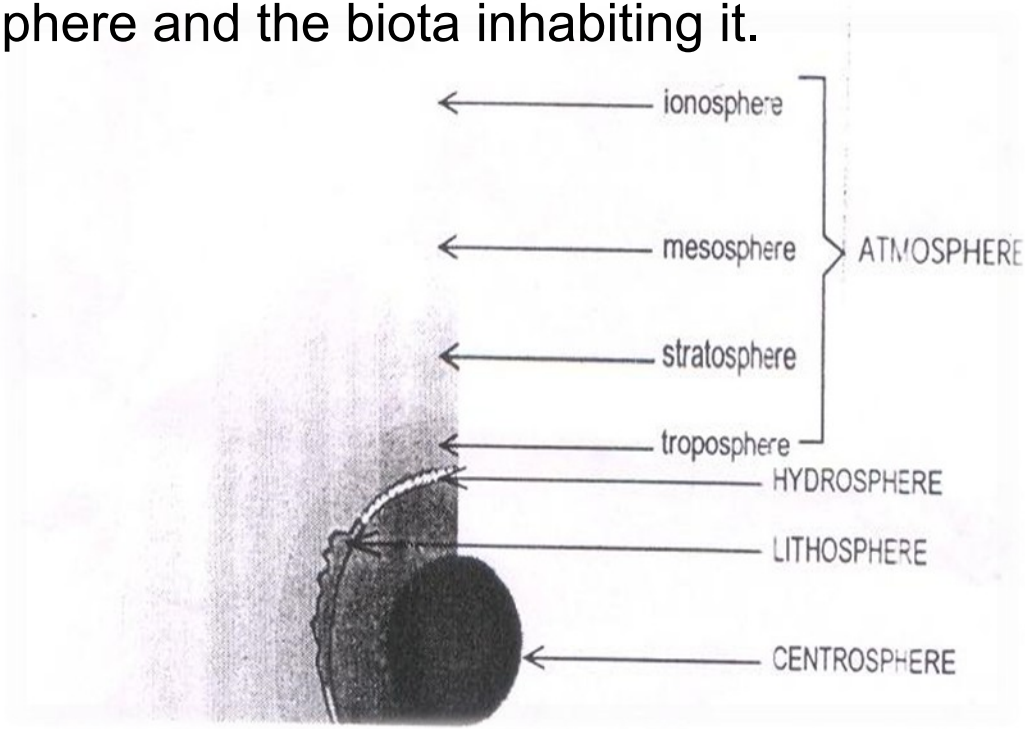
Inhale ⟶ Air (PW)
Eat ⟶ Food (LW)

http://www.yourarticlelibrary.com/essay/environment-and-its-components-explained-with-diagram/44682

Environment and **Earth System**

Earth System

The earth is a complex system consisting of different components; the lithosphere, the hydrosphere, the atmosphere and the biota inhabiting it.



Environment and **Earth System**

Centrosphere

Earth was once a **molten mass** and to this date, the temperature and pressure in the interior of the earth is so high that the molten state still exists there. These are brought to the surface from time to time as **lava**, through **volcanic eruption** or through faults in the earth's crust.

Lithosphere

The lithosphere consists of the **solid crust** on which rest the continental masses comprising of **mountains, valleys and plains**. This is where human beings and other terrestrial animals live. The soil covering the solid masses in most cases is fertile enough to support plant life and in its natural state is covered with vegetation.

Environment and **Earth System**

Hydrosphere

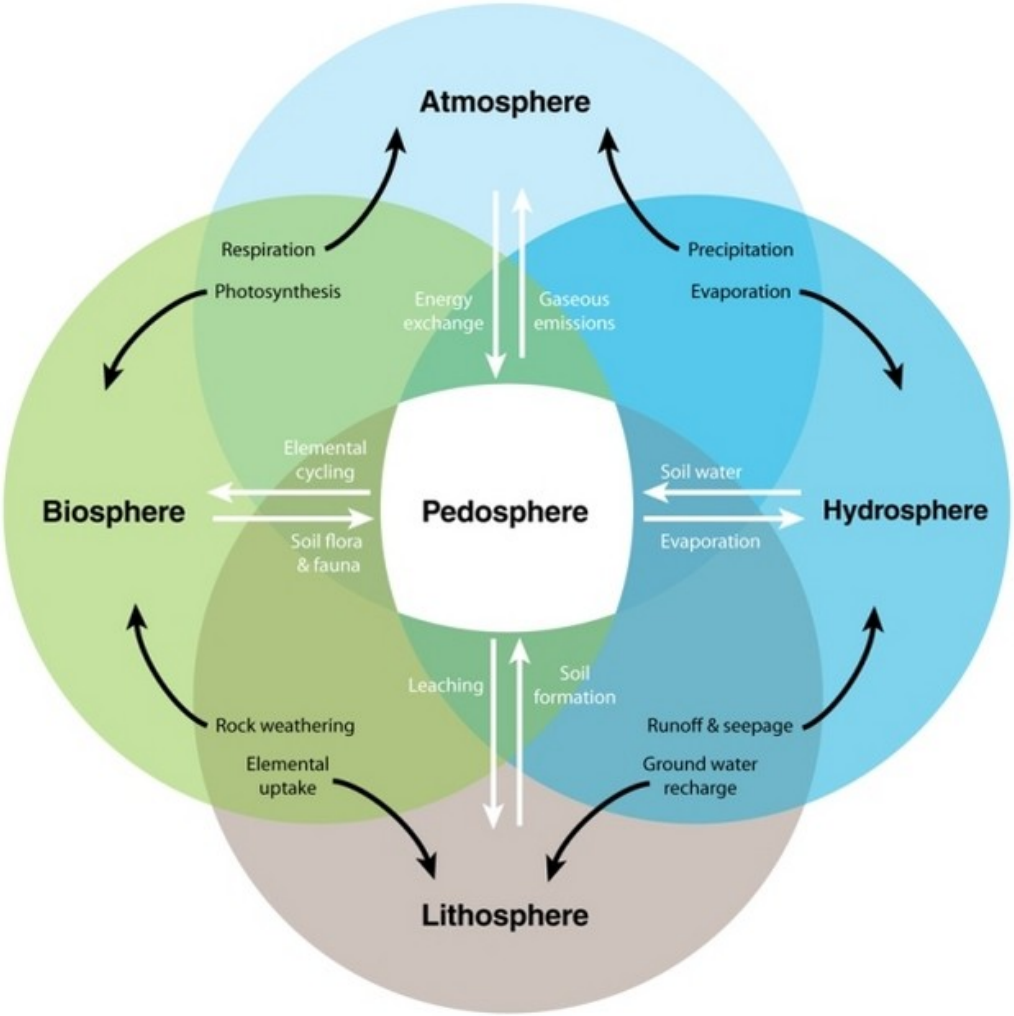
The hydrosphere consists of these oceans, seas and the inland water bodies. **Ninety-seven percent** of the total supply of water on earth are held in the oceans and seas whilst the rest is freshwater. Life on earth would not be possible without water. All the life supporting processes of the earth system are heavily dependent on water.

Atmosphere

The atmosphere is a gaseous mantle enveloping the earth. The main constituents are nitrogen (78%) and Oxygen (21 %). The rest 1 percent is mostly argon, carbon dioxide and some other trace gases.
It consists of different strata called: Troposphere, Stratosphere, Mesosphere and Thermosphere.

Biota: all living organism including Flora and Fauna

Environment and **Earth System**



Ecology and Ecosystem

The study of **living organisms** in the **natural environment**. How they interact with one another and how they interact with their non living environment.

Ecology is the relationship between organism and environment by Hackel and Tansley

Terminologies

Autecology: Individual (Study of RB Tiger Population)

Synecology: Group/community (Study of Forest Ecosystem)

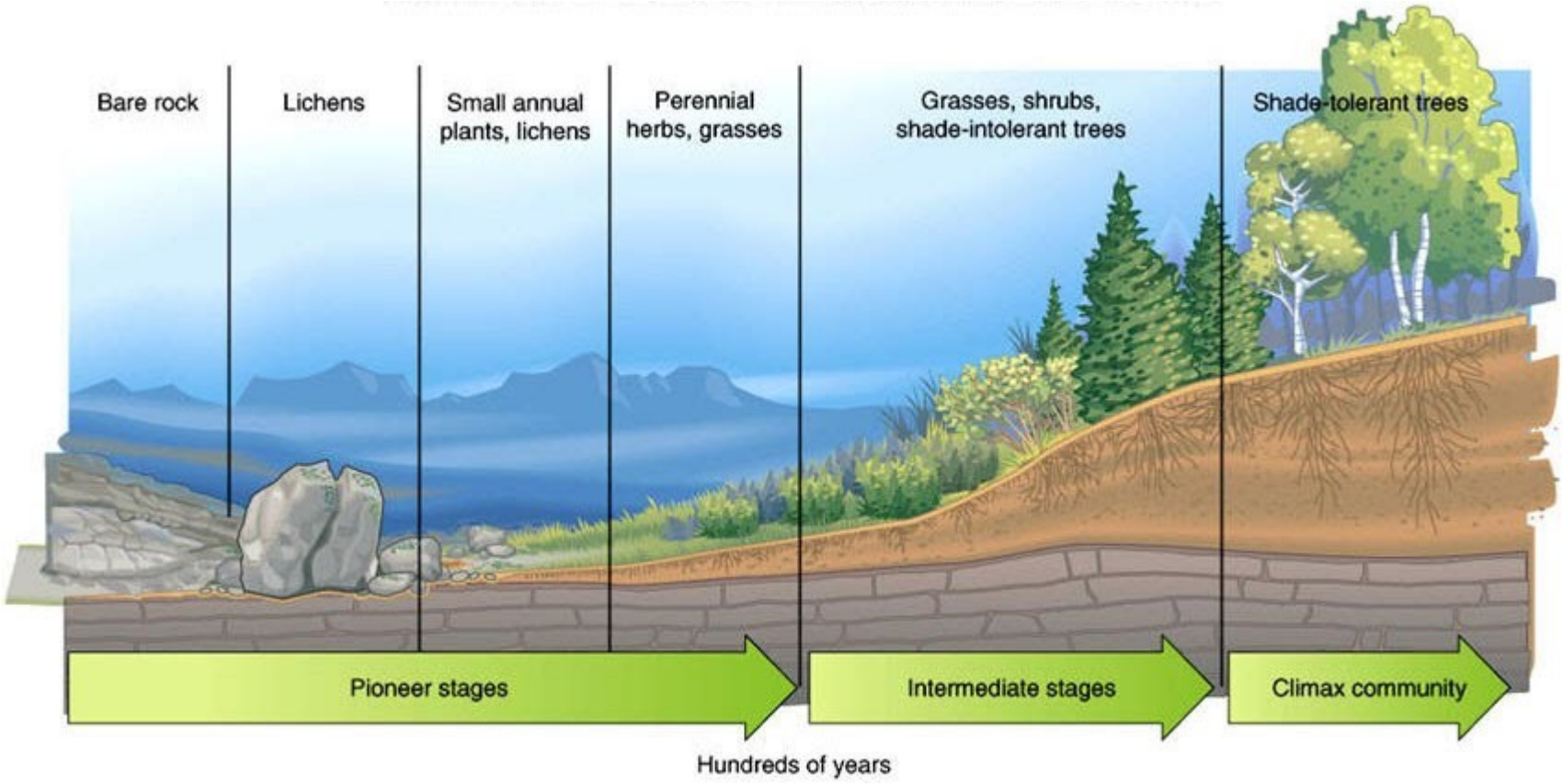
Ecological Succession: **process of change** in the species structure over time

Ecological Niche: Physical + Functional **role of a species** in a community

Ecological Equivalent: Similar niche in different geographical regions

Ecology and Ecosystem

Ecological Succession

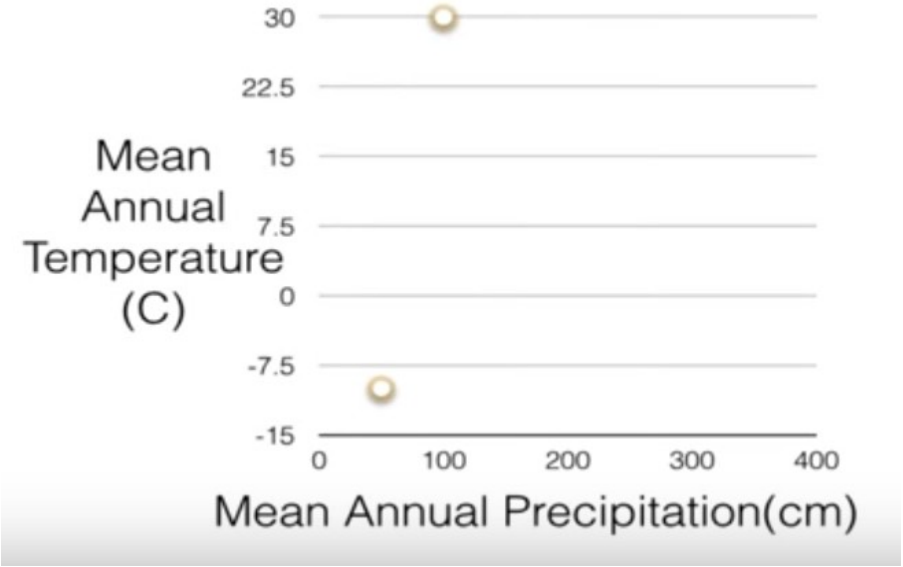


Ecology and Ecosystem

Therms:

- Mega: High, T.rain Forest
- Meso: High+Low, T. Deciduos Forest
- Micro: Low Temperature, Mixed Coniferous
- Hekisto: Extremely low, Alpine

Biomes: very large ecological areas; climate, vegetation, soil



Ecology and Ecosystem

A biome is determined mainly by its climate-like temperature and rainfall.
The earth's land areas are divided into **6 major biomes**.

1. TUNDRA

- Bitterly cold and covered with snow and ice.
- Ground remains permanently frozen (permafrost).
- Only certain small plants such as mosses and grasses can grow.



2. CONIFEROUS FOREST (TAIGA)

- Confers = cone-bearing trees (pines, fir)
- Cold climate
- Dense forests (tree tops block out a lot of sunlight)
- Grasses, small trees cannot grow –only some shrubs, ferns and mosses thrive.
- Animals such as squirrels, moose (sambur), birds, and insects live here.



Ecology and Ecosystem

3. DECIDUOUS (TEMPERATE) FOREST

- Thrive in moderate climates.
- Summers may be hot and winters may be cold.
- Dense forests
- Provides habitats for many animals.



4. TROPICAL RAIN FOREST

- Excellent for plant growth and soil development.
- Plants grow thick and tall.
- Found in areas near the equator.
- Support more plant life and animal species than any other biome.



Ecology and Ecosystem

5. GRASSLANDS

- Mostly grass!
- Temperatures similar to deciduous forest.
- Do not receive much rainfall –enough rain to support grasses but not trees.



6. DESERT

- Very dry; receives very little rainfall.
- Very hot during the day, cold at night.
- Soil is very dry and poor.
- Only a few kinds of plants grow in the desert.
- Very few animals can survive in the desert.



Ecology and Ecosystem

Species

A group of organisms that can breed to produce fully fertile offspring



Great White Pelican *Pelecanus onocrotalus*

Ecology and Ecosystem

Population

A group of organism of the **same species** which live in the same habitat at the same time where they can freely interbreed



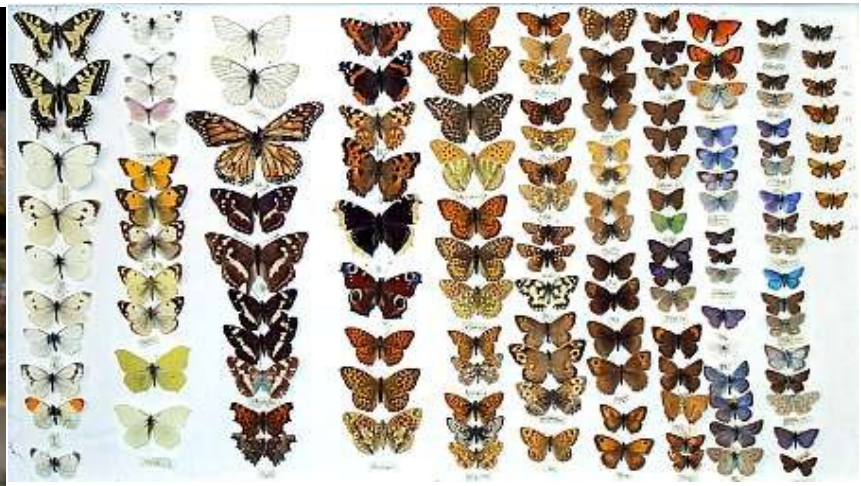
The black-veined white butterfly *(Aporia crataegi)* mating

Ecology and Ecosystem

Biodiversity

The total number of different species in an ecosystem and their relative abundance





Ecology and Ecosystem

The characteristics of the type of environment where an organism normally lives. (e.g. a stoney stream, a deciduous temperate woodland, Bavarian beer mats)

Habitat



Ecology and **Ecosystem**

A unit which consists of all living organisms functioning together in a given area and interacting with the physical environment in a manner, whereby, flow of energy leads to clearly defined biotic structures an cycling of materials between living and non-living parts is know as an ecosystem.

- Ecosystem, the term coined by Tansley
- Smallest structural and functional unit of nature
- **Consisting of both biotic community and the abiotic environment which have close interactions**
- The interaction is conducted by
  energy flow (Solar energy)
  in the system and cycling of materials (**Nutrient cycle**)
- Entire biosphere is considered as a global ecosystem

Ecology and **Ecosystem**

Community + Abiotic environment, interacting



Ecology and **Ecosystem**

Characteristics of Ecosystem

Boundaries are indistinct and over-lapping; could be temporary or permanent; even may be small (drop of water) or large (whole earth)

Types of Ecosystem: on the basis of the origin

1.**Natural** Ecosystem; terrestrial (forest, grassland and desert) and aquatic (pond, lake, wetland, river and estuary).

In aquatic ecosystem, more biodiversity is found in LOTIC ecosystem (running fresh water; stream, river) than in LANTIC ecosystem (Stagnant fresh water; lake, pond, swamp)

2. **Man-made** Ecosystem; crop field and aquarium.

Ecology and **Ecosystem**

Types of Ecosystem: **on the basis of the size**

- Mega Ecosystem: Entire Biosphere
- Macro Ecosystem: Major Biomes like Tropical Rain forest (Amazon)
- Micro Ecosystem: Pond
- Nano Ecosystem: Drop of water

COMPONENTS OF ECOSYSTEM

1. **Biotic Components** includes living component from smallest microbe, fungi to largest animals and plants
2. **Abiotic Component** includes Inorganic substance (carbon, nitrogen, CO_2, H_2O etc.) involved in natural cycles, air, water and other physical factors, organic compounds (Proteins, Carbohydrates, Humus)

Ecology and **Ecosystem**

Among all abiotic components; The important ones are:

- **Temperature**; spatial distribution of species decreases on either side of equator towards the poles and as we go above and below the MSL.
- Water; Life started in water about 3.5 billion years ago.
- Light; photosynthesis
- Soil; Edaphic factors. Based on formation process, it is classified in two types:

Residual soil – Weathering and pedogenesis occur at the same place
Transported soil - Weathering and pedogenesis occur at different places (Colluvial- Brought by gravity, Alluvial- brought by water, Glacial- by snow, Eolian- by wind)

Structure and Function of Ecosystem

Vertical distribution species occupying different levels is called stratification. For instance, trees canopy is top layer of a forest while grasses are at the bottom.

Functions:
a) Productivity b) Decomposition c) Energy Flow and d) Nutrient Cycle

a) Productivity

Biomass: Mass of living biological organism in a given area at a given time

species biomass-mass of one species

community biomass- mass of all species in the community (including microorganisms, plants or animals). It is measured either in terms of dry weight (60-80% weight of water not included) or by measuring organically bound carbon.

Structure and Function of Ecosystem

Primary Production: Total amount of biomass or organic matter produced per unit area. Unit in terms of weight (gm^{-2}) or energy ($kcal/m^2$)

Primary productivity: Amount of biomass produced per unit area over a time period by plants during photosynthesis ($gm^{-2}y^{r-1}/kcalm^{-2}yr^{-1}$)
Secondary Productivity is defined as the rate of formation of new organic matter by consumer

Decomposition
Decomposers break down complex organic matter into inorganic substances like CO_2, water and nutrients and the process is called decomposition.

Structure and Function of Ecosystem

Energy flow

Sun is the only source (with negligible exceptions) of energy for all ecosystems on earth.

Incident solar radiation, if 100, about 50 per cent is Photosynthetically Active Radiation (PAR) from 400-700 nm, which is also the visible spectrum. Out of PAR, **2-10% is actually fixed into organic matter**. And the Unidirectional flow of energy from sun to autotrophs to heterotrophs start.

- Law of thermodynamics are obeyed in this energy flow.
- Energy neither created not destroyed, only transformed.
- Entropy law, transfer of energy leads to loss of energy as heat.

Structure and Function of Ecosystem

Energy is transferred from one **trophic level** to another

- First trophic level (T1) = Producer (Phytoplankton, Grass, Trees)
- Second trophic level (T2) = Primary consumer (Zooplankton, rabbit)
- Third trophic level (T3) = secondary consumer (Birds, fishes)
- Fourth trophic level (T4) = top consumer (Man, Lion)

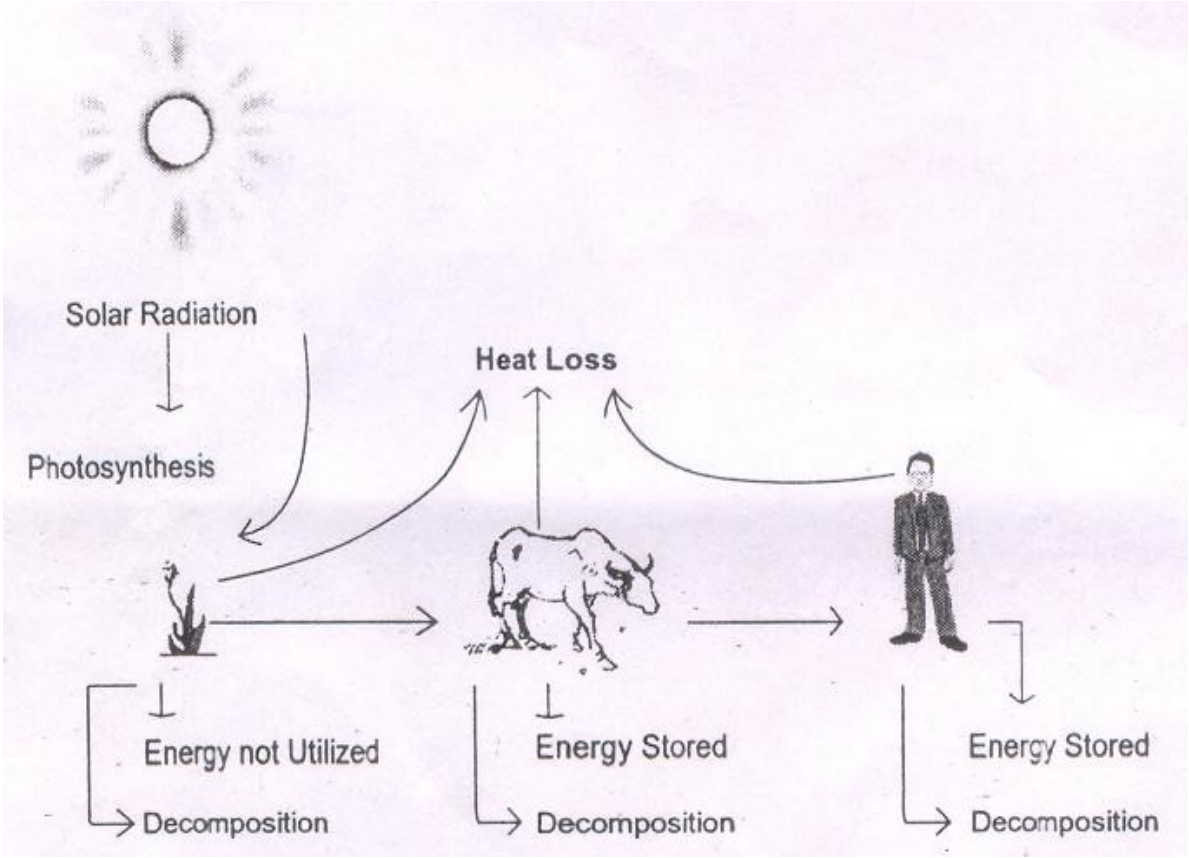
Structure and Function of Ecosystem

Rule of 10% (**10% law of Lindeman**): energy is reduced by 10% during flow to every higher trophic level. Because of this law, the number of trophic levels are usually restricted to 4-5.

Right from the sunlight being trapped by autotrophs into organic material, and further transfer of energy to higher trophic levels. On an avegrage, 10% (2-20%) of energy is transferred while rest 90% (80-98%) is lost in respiratory and other losses. So as the number of trophic levels increase in a food chain, most of the energy is lost.

Structure and Function of Ecosystem

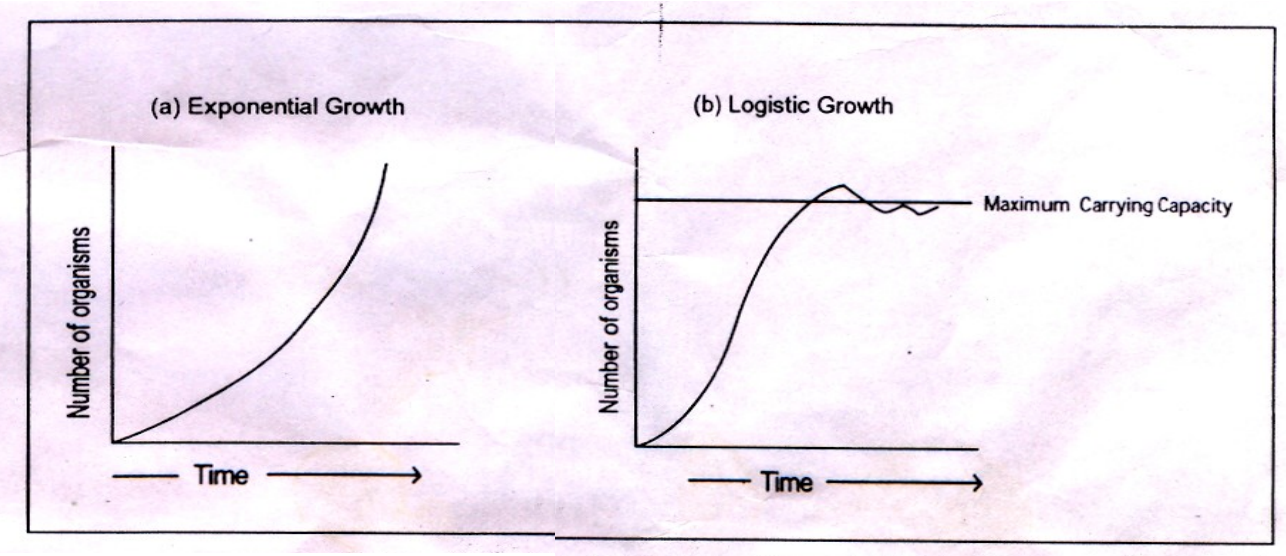
Energy Flow in the Food Chain



Carrying Capacity

Carrying capacity of environment or an ecosystem is the **threshold limit** of use of that system without damaging it.

Every ecosystem has its resources that are used for economic development, for survival and for habitat creation. Environment and ecosystems have got the abilities to recover the loss of its resources by regenerating them over the period of time that are temporary and not exceeding the threshold damage limit.



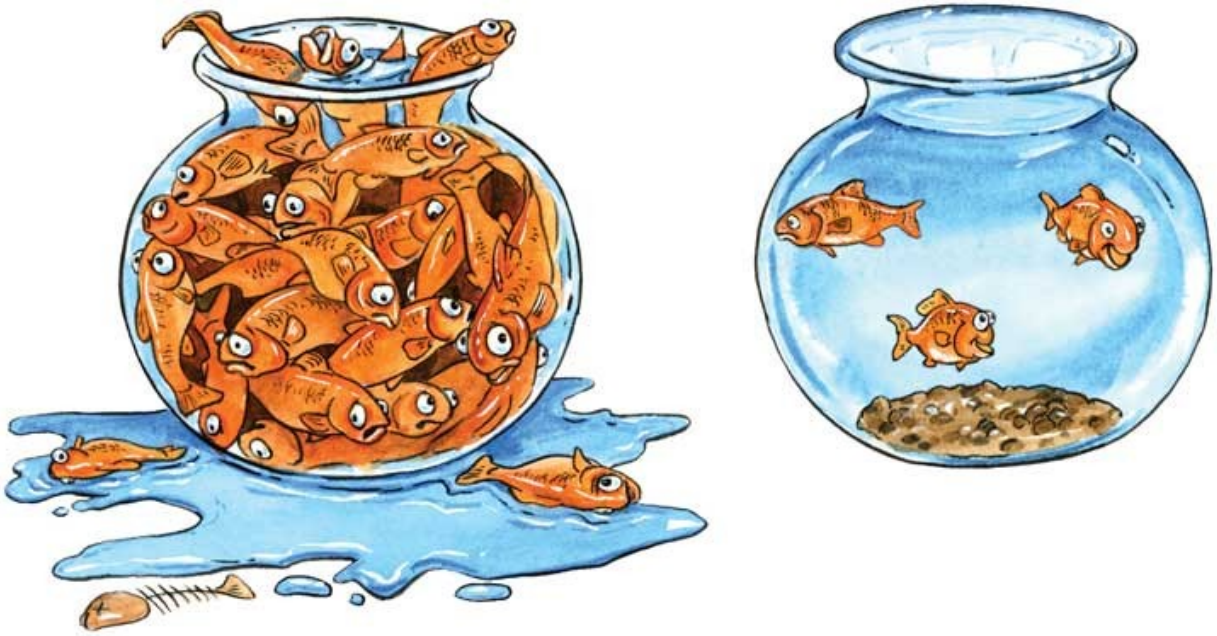
Carrying Capacity

Catton (1986), defined an environment's carrying capacity as its maximum persistently supportable load

Ecological carrying capacity is a measure of the amount of renewable resources in the environment in units of the number of organisms these resources can support. Normally, K is a function of both the species and the environment, and is expected to change only in evolutionary time (Daily and Ehrlich, 1992).



Carrying Capacity



Thanks
Welcoming Q & A